

TABLE IV

ASSOCIATION PARAMETERS FOR PURE COMPONENTS

COMPOUND (GROUP)	HAYDEN (GROUP)	HAYDEN (FITTED)	EXTENDED PITZER ^a
Hydrogen Cyanide	0.12	0.12	0.
Carbon Dioxide ^{b+}	0.16	0.16	0.
Carbon Disulfide ⁺	0.34	0.34	0.
Nitrous Oxide ^{c+}	0.09	0.09	0.
Phosphine	0.05	---	0.30
Dichloromethane	0.	0.	0.32
Trifluoromethane	0.	0.	0.20
Difluorochloromethane	0.	0.	0.50
Fluorodichloromethane	0.	0.	0.30
Trichloromethane	0.	0.	0.28
Water (Hydroxyl)	1.55	1.70	0.
Methanol (Hydroxyl)	1.55	1.63	1.10*
Ethanol (Hydroxyl)	1.55	1.40	0.70*
1-propanol (Hydroxyl)	1.55	1.40	0.57
2-propanol (Hydroxyl)	1.55	1.32	0.65
1-Butanol (Hydroxyl)	1.55	2.2	0.45
2-Butanol (Hydroxyl)	1.55	1.75	0.58
2-Methyl-1-Propanol (Hydroxyl)	1.55	1.90	0.49
2-Methyl-2-Propanol (Hydroxyl)	1.55	1.00	0.54
Phenol (Phenyl Hydroxyl)	0.32	0.32	0.55
Methoxy Methane	0.0	0.0	0.55
Methoxy Ethane	0.	0.	0.20*
Ethoxy Ethane	0.	0.	0.28

TABLE IV

2

COMPOUND (GROUP)	HAYDEN (GROUP)	HAYDEN (FITTED)	EXTENDED PITZER ^a
Methoxy Propane	0.	0.	0.18*
Methoxy Pentane	0.	0.	0.14*
Methoxy Benzene	0.0	0.0	0.30*
Furan	0.13	0.13	0.00
Acetaldehyde (Aldehyde)	0.58	0.58	0.
Dimethyl ketone (Ketone)	0.90	0.90	0.
Methyl ethyl ketone (Ketone)	0.90	0.90	0.30
Methyl-n-propyl ketone (Ketone)	0.90	0.60	0.17*
Diethyl ketone	0.90	1.50	0.28*
Methyl formate (Formate)	0.20	0.35	0.45*
Methyl acetate (Ester)	0.70	0.85	0.62
Ethyl formate (Formate)	0.20	0.15	0.36
Methyl propionate (Ester)	0.70	0.80	0.49
Ethyl Acetate (Ester)	0.70	0.53	0.50
Propyl Formate (Formate)	0.20	0.23	0.39
Formic Acid (Organic Acid)	4.5	--	--
Acetic Acid (Organic Acid)	4.5	--	--
Propionic Acid (Organic Acid)	4.5	--	--
Butyric Acid (Organic Acid)	4.5	--	--
Heptanoic Acid (Organic Acid)	4.5	--	--
Ammonia	0.20	0.0	0.0
Methyl Amine (Amine)	0.20	0.13	0.72
Dimethyl Amine (Amine)	0.20	0.17	0.56
Ethyl Amine (Amine)	0.20	0.30	0.62
Trimethylamine (Amine)	0.20	0.06	0.20
Diethylamine (Amine)	0.20	0.21	0.25*

TABLE IV

3

COMPOUND (GROUP)	HAYDEN (GROUP)	HAYDEN (FITTED)	EXTENDED PITZER ^a
Pyro1lidine(Amine)	0.20	0.80	0.42 [*]
Pyridine (Amine)	0.20	0.0	0.2
α -Picoline (Amine)	0.2	0.0	0.33 [*]
β -Picoline (Amine)	0.2	0.35	0.37 [*]
γ -Picoline (Amine)	0.2	0.1	0.30 [*]
Acetonitrile (Nitrile)	1.65	1.65	0.0
Nitromethane (Nitro)	1.66	1.66	0.07 [*]
Hydrogensulfide (Thiol)	0.15	0.0	0.60 [*]
Ethanethiol (Thiol)	0.15	0.05	0.31
Propanethiol (Thiol)	0.15	0.25	0.35 [*]
2-Propanethiol (Thiol)	0.15	0.11	0.32 [*]
1-Butanethiol (Thiol)	0.15	0.85	0.41 [*]
2 Methyl-1-Butanethiol (Thiol)	0.15	0.17	0.36 [*]
2 Methyl-2-Butanethiol (Thiol)	0.15	0.03	0.32 [*]
1 Pentanethiol (Thiol)	0.15	0.13	0.29 [*]
Benzenethiol (Thiol)	0.15	0.30	0.31 [*]
Dimethyl Sulfide	0.	0.	0.31 [*]
Methyl Ethyl Sulfide	0.	0.	0.38 [*]
Methyl-n-propyl Sulfide	0.	0.	0.31 [*]
Methyl-i-propyl Sulfide	0.	0.	0.31 [*]
Diethyl Sulfide	0.	0.	0.34 [*]
Thiophene	0.	0.	0.18 [*]

^a J. P. O'Connell and J. M. Prausnitz, Ind. Eng. Chem. Proc. Des. Devel., **6**, 245 (1967) *Revised value.

^b Pseudocritical constants for present correlation $T_c = 241^\circ\text{K}$, $P_c = 53.1\text{atm}$, $V_c = 108\text{ cm}^3/\text{gmol}$. Also recommended for ethyne.

^c Pseudo critical constants for present correlation $T_c = 277^\circ\text{K}$, $P_c = 63.4\text{ atm}$, $V_c = 104\text{ cm}^3/\text{gmol}$.

+ Association constant adopted to account for large quadrupole moment.

TABLE V

SOLVATION PARAMETERS FOR MIXED SYSTEMS

<u>SYSTEM</u>	<u>HAYDEN</u> ^a	<u>EXTENDED</u> <u>PITZER</u> ^b	<u>NOTHNAGEL</u> <u>ET AL.</u> ^c	<u>TSONOPOULOS</u> ^d
Hydrogen Chloride - Water	1.38	2.20	550.	-0.26
Carbon Dioxide - Water	0.32 ^e	0.15	130	0.14
Carbon Dioxide - Methanol	0.32 ^e	--	100	0.01
Carbon Dioxide - Nitromethane	0.16	--	90	0.18
Carbon Disulfide - Acetone	0.13	0.	50	0.1
Nitrous Oxide - Water	0.17 ^e	0.15	100	0.16
Nitrous Oxide - Methanol	0.17 ^e	--	60	0.13
Sulfur Dioxide - Methoxybutane	0.58	1.03	100.	-0.03
Ammonia - Ethylene	0.20	0.75	275.	-0.1
Ethylene - Methanol	0.12	--	120.	0.1
Benzene - Chloroform	0.12	0.04	63.1	-0.01
Benzene - Acetone	0.50	0.02	156.	0.03
Benzene - Nitromethane	1.00	0.00	155	0.0
Chloromethane - Acetone	0.20	0.05	40.6	0.01
Chloromethane - Acetone	0.92	0.33 [*]	192	-0.04
Chloroform - Diethylether	0.95	0.38 [*]	205	-0.08
Chloroform - Acetone	1.26	0.55 [*]	175	-0.05
Chloroform - Methyl Formate	1.20 ^e	0.32	121.4	-0.06
Chloroform - Methyl Acetate	1.65 ^e	0.47	250.	-0.15
Chloroform - Ethyl Acetate	1.65 ^e	0.47	263	-0.14
Chloroform - Propyl Formate	1.20 ^e	0.35	215	-0.10
Chloroform - Diethylamine	1.57	0.62	391.6	-0.20
Acetone - Nitromethane	1.63	0.5	110	-0.05
Acetaldehyde - Acetonitrile	2.5	3.8	471	-0.28

TABLE V (con't)

<u>SYSTEM</u>	<u>HAYDEN^a</u>	<u>EXTENDED PITZER^b</u>	<u>NOTHNAGEL ET AL.^c</u>	<u>TSONOPOULOS^d</u>
Methane - Tetrafluoromethane	-0.0154 ⁺	-1.	0.	0.1
Methane - Perfluorobutane	-0.0321 ⁺	0.	0.	0.1
Ethane - Tetrafluoromethane	-0.028 ⁺	-1.	0.	0.1
Ethane - Perfluorobutane	-0.062 ⁺	-0.25	0.	0.1
n-Butane - Tetrafluorobutane	-0.044 ⁺	-0.7	0.	0.1
n-Butane - Perfluorobutane	-0.111 ⁺	-0.2	-600.	0.1
Neopentane - Tetrafluoromethane	-0.046 ⁺	0.	0.	0.1
n-Hexane - Tetrafluoromethane	-0.058 ⁺	-0.37	-300.	0.1
n-Hexane - Perfluorobutane	-0.172 ⁺	-0.23	-700.	0.1

^aWhere one component associates, individual component η used as input data.

^bJ. P. O'Connell and J. M. Prausnitz, Ind. Eng. Chem. Process, Des. Develop., 6, 245 (1967). Values reported here should be added to that obtained from equation (18) of that paper.

* Indicates revised value from Table V of that paper.

^cK.-H. Nothnagel, D. S. Abrams, and J. M. Prausnitz, Ind. Eng. Chem. Process. Des. Develop., 12, 25 (1973). Values reported here are $-\Delta H_{12}^*$, cal/mol of Table III of that paper.

^dC. Tsonopoulos, AIChEJ, 20, 263 (1974). Values reported here are values of k_{12} used in Eq (18) of that paper.

^e Used for more than one pair with same solvating groups, e.g. CO₂-hydroxyl, CHCl₃-formate, etc.

⁺Calculated from $\eta = 3 \times 10^{-5}(\epsilon/k) - 7 \times 10^{-7}(\epsilon/k)^2 - 3 \times 10^{-23}(\epsilon/k)^8$ where ϵ/k from Equation (32)

8-246-1193

SECURITY

USAGE MAIN PROGRAM

TC	CRITICAL TEMPERATURE, DEG. K
PC	CRITICAL PRESSURE, ATM.
R	MEAN RADIUS OF GYRATION, ANGSTROMS
MU	DIPOLE MOMENT, DEBYE (1 E=18 ESU)
ETA	ASSOCIATION PARAMETER (PURE COMPONENT)
NAME	ALPHANUMERIC NAME OF SUBSTANCE (MAX 8 CHARACTERS)
ETA12	SOLVATION PARAMETER (UNLIKE INTERACTIONS)
T	ABSOLUTE TEMPERATURE, DEG. K

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1ST CARD TC,PC,R,MU,ETA,NAME      (FORMAT 100)
2ND CARD BLANK FOR PURE COMPONENTS
      REPEAT 1ST CARD FOR 2ND COMPONENT
      IN MIXTURE
3RD CARD (USED ONLY FOR MIXTURE) ETA12 (FORMAT 100)
4TH CARD T                          (FORMAT100)
      REPEAT FOR ALL DESIRED TEMPERATURES
      BLANK FOR NEW SYSTEM
      NEGATIVE TO STOP

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1      REAL MU,MUSTAR,MUSTRP
2      DIMENSION TC(2),PC(2),R(2),MU(2),ETA(2),NAME(2,2),OMEGA(2),
1      SIGMA3(2),EPSOK(2),SIGMA(2)

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3      5 READ (5,100) TC(1),PC(1),R(1),MU(1),ETA(1),NAME(1,1),NAME(2,1)
4      NUMCOM=2
5      READ (5,100) TC(2),PC(2),R(2),MU(2),ETA(2),NAME(1,2),NAME(2,2)
6      IF(TC(2).EQ.0.) NUMCOM=1
7      IF(NUMCOM.EQ.2) READ(5,100) ETA12

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8 DD 10 I=1,NUMCOM
9 OMEGA(I)=0.006*R(I)+0.02087*R(I)**2=0.00136*R(I)**3
10 EPSOK(I)=TC(I)*(748+0.91*OMEGA(I)-0.4*ETA(I)/(2.+20.*OMEGA(I)))
11 SIGMA3(I)=(2.44*OMEGA(I))**.33*TC(I)/PC(I)
12 SIGMA(I)=SIGMA3(I)**(1./3.)
13 IF(MU(I).LT.1.45) GO TO 10
14 POTN=16.+400.*OMEGA(I)
15 POWR1=POTN/(POTN-6.)
16 POWR2=POWR1/POTN
17 CONST=(2.882-1.887*OMEGA(I)/(0.03+OMEGA(I)))*5.723E-08
18 XI=MU(I)**4/(CONST*EPSOK(I)*(SIGMA3(I)**2)*TC(I))
19 EPSOK(I)=EPSOK(I)*(1.-XI*POWR1+POWR1*(1.+POWR1)*XI**2/2.)
20 SIGMA3(I)=SIGMA3(I)*(1.+3.*XI*POWR2)
21 10 CONTINUE
22 IF(NUMCOM,EQ,2) GO TO 20

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23 EPSK12=EPSOK(1)
24 SIG123=SIGMA3(1)
25 OMEG12=OMEGA(1)
26 MU(2)=MU(1).
27 ETA12=ETA(1)
28 GO TO 30

C PARAMETERS FOR MIXTURE CALCULATION

Computer Program for Calculating Pure and Cross Virial
Coefficients Using Correlation of Hayden and O'Connell

PART VI

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C
C NONPOLAR-NONPOLAR MIXTURES
29 20 EPSK12=0.6/(1./EPSOK(1)+1./EPSOK(2))+SQRT(EPSOK(1)*EPSOK(2))*0.7
30 SIG123=SQRT(SIGMA3(1)*SIGMA3(2))
31 OMEG12=0.5*(OMEGA(1)+OMEGA(2))
32 30 IF(MU(1)*MU(2).NE.0.) GO TO 40
C
C POLAR-NONPOLAR MIXTURES
33 MUSTAR=0.
34 MUSTRP=0.
35 IF(MU(1)+MU(2).LT.2.) GO TO 50
36 XI12=(MU(1)**2*(EPSOK(2)**2*SIGMA3(2))**(.1/3.)*SIGMA3(2)+
1 MU(2)**2*(EPSOK(1)**2*SIGMA3(1))**(.1/3.)*SIGMA3(1))/(EPSK12
2 *SIG123**2)
37 POTN12=16.+400.*OMEG12
38 POWR1=POTN12/(POTN12-6.)
39 POWR2=POWR1/POTN12
40 EPSK12=EPSK12*(1.+XI12*POWR1)
41 SIG123=SIG123*(1.-3.*XI12*POWR2)
42 GO TO 50
C
C POLAR-POLAR MIXTURES
43 40 MUSTAR=MU(1)*MU(2)/(1.3805E-04*EPSK12*SIG123)
44 MUSTRP=MUSTAR-0.25
45 IF(MUSTAR.LT.0.25) MUSTRP=0.
46 IF(MUSTAR.LT.0.04) MUSTRP=MUSTAR
C
C FINISH CALCULATION OF PARAMETERS
47 50 DELTAS=650./(EPSK12+300.)
48 IF(ETA12.GT.4.) DELTAS=42800./(22400.+EPSK12)
49 A=-0.3-0.05*MUSTAR
50 DELTAH=1.99+0.2*MUSTAR**2
51 B012=1.2618*SIG123
52 SIGM12=SIG123**(.1/3.)
C
C OUTPUT PROPERTIES AND PARAMETERS
53 WRITE(6,1000) (NAME(1,I),NAME(2,I),TC(I),PC(I),R(I),MU(I),ETA(I),
1 EPSOK(I),SIGMA(I),I=1,NUMCOM)
54 IF(NUMCOM.EQ.2) WRITE(6,1001) EPSK12,SIGM12,ETA12
55 WRITE(6,1002)
C
C INPUT TEMPERATURE
56 60 READ(5,100) T
57 IF(T) 80,5,65
58 65 BCHEM=0.
C
C CALCULATION OF PHYSICAL CONTRIBUTIONS TO SECOND VIRIAL COEFFICIENT
59 TSTAR=T/EPSK12
60 TSTRP =1./TSTAR-OMEG12*1.6
61 BFREE=(.94-1.47*TSTRP-0.85*TSTRP**2+1.015*TSTRP**3)-MUSTRP*(
1 10.75-3.*TSTRP+2.1*TSTRP**2+2.1*TSTRP**3)*B012
62 B=BFREE+(B012*A*EXP(DELTAM/TSTAR))
63 IF(ETA12.EQ.0.) GO TO 70
C
C CALCULATION OF CHEMICAL CONTRIBUTIONS (ASSOCIATION OR SOLVATION)
64 BCHEM=B012*EXP(ETA12*(DELTAS-4.27))*(1.-EXP(ETA12*1500./T))
65 B=B+BCHEM
C
C OUTPUT OF VIRIAL COEFFICIENT
66 70 WRITE(6,1003) T,B,BFREE,BCHEM
67 GO TO 60
68 80 STOP
69 100 FORMAT(5F10.0,2A4)
70 1000 FORMAT('1 CALCULATION OF SECOND VIRIAL COEFFICIENTS FROM CORRELATIO
1 N OF HAYDEN AND O'CONNELL',/3X,2A4,' TC=',F8.3,'K PC=',F8.2,'ATM
2 R=',F6.3,'A. MU=',F6.2,'DERBY ETA=',F6.2,' E/K=',F8.2,'K SIGMA
3 =',F6.3,'A')
71 1001 FORMAT('1 MIXTURE PARAMETERS (F/K)12=',F8.2,'K SIGMA12=',F6.3,
1 'A ETA12=',F6.2)
72 1002 FORMAT('1 T,K BTOTAL,CC/GMOL BFREE,CC/GMOL BCHEM,CC/GMOL')
73 1003 FORMAT(F7.2,F10.2,2F16.2)
74 END

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8-26-1994

8-216-195

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C*****
C
C          *****
C          *RGYRA*
C          *****
C
C    PURPOSE  TO CALCULATE MEAN RADIUS OF GYRATION
C              AND MOMENTS OF INERTIA OF MOLECULES FROM
C              ATOMIC MASSES AND COORDINATES
C
C    USAGE    MAIN PROGRAM
C
C    DESCRIPTION OF PARAMETERS
C      L      INDEX OF LINEARITY
C              L=0 FOR LINEAR
C              L=1 FOR NONLINEAR
C      NOAT    NUMBER OF ATOMS
C      NAME    ALPHANUMERIC NAME OF MOLECULE
C              (MAXIMUM OF 32 CHARACTERS)
C      W(I)    ATOMIC MASS OF ATOMS
C      COR(I,3) CARTESIAN COORDINATES OF ATOMS
C
C    DATA INPUT
C      1ST CARD  L,NOAT,NAME          (FORMAT 200)
C      2ND CARDS W,COR(X),COR(Y),COR(Z) (FORMAT 201)
C              TO STOP, PUT BLANK IN PLACE OF 1ST CARD, OMIT 2ND CARDS
C*****
1  DIMENSION NAME(8),COR(500,3),W(500),CM(3),RT(500),X(500,3),
   1DNER(3,3)
C
C      READ 1ST CARD
C
2  1 READ(5,200) L, NOAT,(NAME(I),I=1,8)
3  200 FORMAT(2I4,8A4)
4  IF(NOAT.LE.0) STOP
C
C      READ 2ND CARDS
C
5  READ(5,201) (W(I),(COR(I,J),J=1,3),I=1,NOAT)
6  201 FORMAT(4F10,4)
7  WRITE(6,300) (NAME(I),I=1,8)
8  300 FORMAT('1ATOM COORDINATES FOR',1X,8A4/'0ATOM NO.',6X,'MASS',9X,
   1'X',9X,'Y',9X,'Z')
9  WRITE(6,301) (I,W(I),(COR(I,J),J=1,3),I=1,NOAT)
10 301 FORMAT(3X,I4,2X,F10,4,1X,F10,4,1X,F10,4,1X,F10,4)
C
C      CALCULATE MOLECULAR MASS
C
11 168 WT=0.0
12 DO 170 I=1,NOAT
13 170 WT=WT+W(I)
C
C      FIND CENTER OF MASS
C
14 DO 180 M=1,3
15 CM(M)=0.0
16 DO 179 I=1,NOAT
17 179 CM(M)=CM(M)+W(I)*COR(I,M)
18 180 CM(M)=CM(M)/ WT
C
C      CALCULATE MOMENTS OF INERTIA
C
19 DO 185 I=1,NOAT
20 RT(I)=0.0
21 DO 185 M=1,3
22 X(I,M)=COR(I,M)+CM(M)
23 185 RT(I)=RT(I)+(X(I,M))**2
24 DO 190 I=1,3
25 DO 190 J=1,3
26 DNER(I,J)=0.0
27 DO 190 K=1,NOAT
28 IF(I-J)189,187,189
29 187 DNER(I,J)=DNER(I,J)+W(K)*(RT(K)-X(K,I)*X(K,I))
30 GO TO 190

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Computer Program for Calculating
Mean Radii of Gyration

PART VII

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31 189 DNER(I,J)=DNER(I,J) -W(K)* X(K,1)*X(K,J)
32 190 CONTINUE
33 IF(L.EQ.0) GO TO 2
C
C      NONLINEAR MOLECULES
C
34 DET=DNER(1,1)*DNER(2,2)*DNER(3,3) +DNER(1,2)*DNER(2,3)*DNER(3,1)
1+DNER(1,3)*DNER(3,2)*DNER(2,1) -DNER(3,1)*DNER(2,2)*DNER(1,3)
2-DNER(2,1)*DNER(1,2)*DNER(3,3) -DNER(1,1)*DNER(2,3)*DNER(3,2)
35 ADET=ABS(DET)
36 R=SQRT((2.*3.14159*(ADET**((1./3.)))/#T)
37 GO TO 4
C
C      LINEAR MOLECULES
C
38 2 IF(DNER(1,1).EQ.0.) GO TO 5
39 IF(DNER(2,2).EQ.0.) GO TO 6
40 DET=DNER(1,1)*DNER(2,2)
41 GO TO 3
42 5 DET=DNER(2,2)*DNER(3,3)
43 GO TO 3
44 6 DET=DNER(1,1)*DNER(3,3)
45 3 ADET=ABS(DET)
46 R=SQRT((SQRT(ADET)/#T)
47 WRITE (6,303)
48 303 FORMAT(// ' LINEAR MOLECULE')
C
C      PRINTOUT
C
49 4 WRITE(6,302) R,((DNER(I,J),J=1,3),I=1,3)
50 302 FORMAT(// ' MEAN RADIUS OF GYRATION=',F10.4//
1' MOMENT OF INERTIA TENSOR'/(1X,3F10.4))
51 GO TO 1
52 END

$DATA

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CART190
CART191

8-216-m96